

High-Performance Chemigenetic Potassium Ion Indicator

Dazhou Cheng, Zhenlin Ouyang, Xiaoyu He, Yusuke Nasu, Yurong Wen, Takuya Terai,* and Robert E. Campbell*

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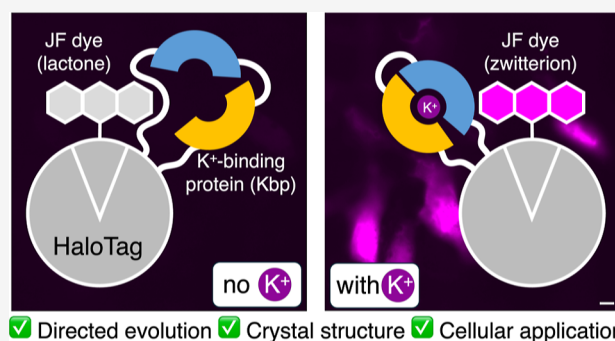


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ABSTRACT: Potassium ion (K^+) is the most abundant metal ion in cells and plays an indispensable role in practically all biological systems. Although there have been reports of both synthetic and genetically encoded fluorescent K^+ indicators, there remains a need for an indicator that is genetically targetable, has high specificity for K^+ versus Na^+ , and has a high fluorescent response in the red to far-red wavelength range. Here, we introduce a series of chemigenetic K^+ indicators, designated as the HaloKbp1 series, based on the bacterial K^+ -binding protein (Kbp) inserted into HaloTag7 self-labeled with environmentally sensitive rhodamine derivatives. This series of high-performance indicators features high brightness in the red to far-red region, large intensimetric fluorescence changes, and a range of K_d values. We demonstrate that they are suitable for the detection of physiologically relevant K^+ concentration changes such as those that result from the Ca^{2+} -dependent activation of the BK potassium channel.



INTRODUCTION

Potassium ion (K^+) is one of the most abundant and important metal ions in biological systems.¹ Generally, K^+ concentration inside cells is about 140–150 mM, while that in the extracellular fluid is about 3.5–5 mM.² This gradient is essential for various physiological activities, ranging from the maintenance of membrane potential in prokaryotes³ to the proper function of the cardiovascular⁴ and immune systems⁵ in humans. Disruption of K^+ homeostasis is related to a variety of diseases of the kidney,⁶ and certain channel proteins related to K^+ uptake and release are drug targets.⁷

Fluorescent indicators are powerful tools to visualize the spatiotemporal dynamics of K^+ in living cells and tissues. The development of synthetic K^+ indicators that are composed of a fluorescent π -conjugation system and a crown ether (or cryptand)-based chelator has a long history dating back to the 1980s. Examples include PBFI,⁸ TAC-Red,⁹ and RPS-1.¹⁰ While these indicators are associated with some advantages, such as generally bright fluorescence and ease of use, they are also associated with disadvantages such as poor membrane permeability, the inability to be targeted to specific tissues or subcellular organelles, and suboptimal specificity for K^+ versus Na^+ . Recently, single fluorescent protein (FP)-based genetically encoded K^+ indicators (also known as sensors or biosensors) based on green FP and a K^+ -binding protein¹¹ (Kbp) have been reported, including the GINKO series^{12,13} and KRaION series.¹⁴ The *Escherichia coli* Kbp protein is composed of an N-terminal BON (bacterial OsmY and nodulation) domain and a C-terminal LysM (lysine motif)

domain (Figure 1a, left) and undergoes a large conformational change upon K^+ binding.¹¹ While the FP-based indicators have proven to be applicable in living cells and provide robust fluorescence changes, they display shorter emission wavelengths (no longer than ~550 nm) and lower molecular brightness than many synthetic indicators. Genetically encoded indicators based on red-shifted FPs have been reported for some targets,^{15–17} but to date there have been no reports of highly optimized red-shifted K^+ indicators.

To combine the strengths of synthetic and genetically encoded indicators, a new class of fluorescent chemigenetic indicators was developed. The earliest such examples were calcium ion (Ca^{2+}) indicators that were created by attaching an environmentally sensitive synthetic fluorophore ligand to a protein framework consisting of a self-labeling protein and a genetically fused Ca^{2+} -binding domain. The first example of such was based on the intermolecular splitFAST self-labeling tag and a rhodamine-based fluorophore.¹⁸ Later examples included rHCaMP¹⁹ and HaloCaMP,²⁰ which used the HaloTag7 (ref 21) self-labeling protein and provided higher dynamic ranges and more red-shifted fluorescence wavelengths. One of the most recent examples is WHaloCaMP,²²

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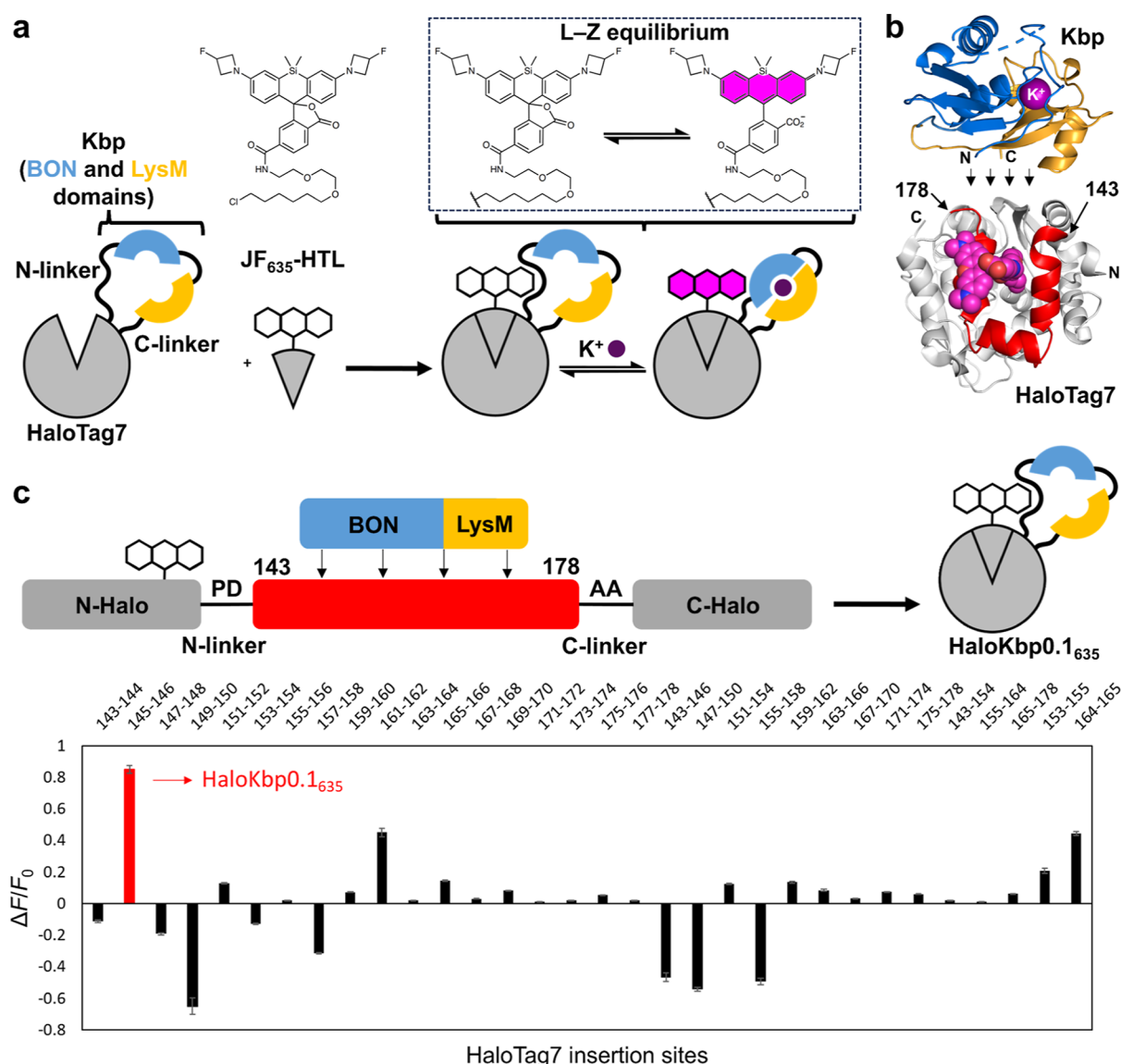


Figure 1. Overall design of HaloKbp indicators and prototype screening. (a) Schematic representation of the HaloKbp indicator structure and response. Based on the precedent of HaloCaMP, the K⁺-binding-dependent conformational change of Kbp could change the environment of the JF₆₃₅-HTL fluorophore and shift the L–Z equilibrium. (b) Crystal structure of the Kbp domain of GINKO1 (PDB ID: 7VCM)¹³ and HaloTag7 (PDB ID: 6U32)²⁰ bound to tetramethylrhodamine (TMR)-HTL. The bound TMR-HTL is shown in atomic sphere representation, and the region explored for insertion of Kbp (residues 143–178) is shown in red. (c) To produce 32 HaloKbp variants, we substituted Kbp in place of different combinations of amino acids between residues 143–178 of HaloTag7. The chart shows the K⁺-dependent $\Delta F/F_0$ when conjugated with JF₆₃₅-HTL for all 32 variants ($n = 3$, mean $\pm$ s.d.). The variant with residues 145–146 of HaloTag7 replaced by Kbp exhibited the largest increase in fluorescence and was designated as HaloKbp0.1₆₃₅.

which was designed to use fluorophores with improved biodistribution for in vivo applications. In all of these examples, the fluorescence color or intensity is altered due to changes in the chromophore environment that result from Ca²⁺-binding-induced protein conformational changes. Similar designs of indicators, based on HaloTag7 and the bright far-red Janelia Fluor 635 HaloTag ligand (JF₆₃₅-HTL),^{20,23} have now been reported for kinase activity²⁴ and H₂O₂.²⁵ In addition, indicators for glutamate, K⁺, and ATP have been reported based on the FAST self-labeling protein and green-, yellow-, and red-emitting rhodamine-based fluorophores.²⁶ Of particular relevance to this work, the FAST-based green-emitting K⁺ indicator, K⁺-FAST-5.1, has an apparent dissociation constant (K_d) for K⁺ of 2 mM and a 2.2-fold increase in fluorescence upon binding.²⁶ This affinity for K⁺ is too high for intracellular

imaging applications, and the fluorescence increase upon binding is less than what can be routinely achieved with highly optimized FP-based indicators.^{13,27,28} Application of K⁺-FAST-5.1 in living cells was not demonstrated.

Inspired by the examples of chemigenetic indicators described above, we developed a series of chemigenetic K⁺ indicators called HaloKbp1_X (where X corresponds to the wavelength of the conjugated Janelia Fluor dye). The protein scaffold of this indicator is based on the insertion of the Kbp protein into a self-labeling HaloTag7 protein, which is conjugated by a series of rhodamine derivative dyes with more red-shifted wavelengths than typical FPs.²³ By systematic screening of insertion sites and extensive protein engineering-based optimization, we developed indicators with up to a 30-fold increase in fluorescence upon binding to K⁺ and a range of

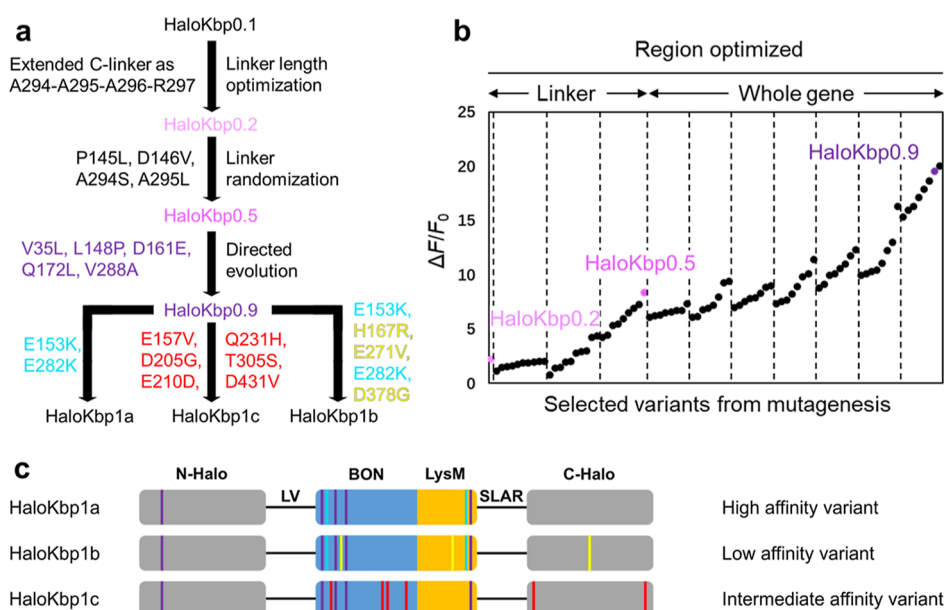


Figure 2. Development of the HaloKbp1 series, starting from HaloKbp0.1. (a) Lineage of HaloKbp variants derived from HaloKbp0.1. (b) $\Delta F/F_0$ of the top variants through rounds of linker optimization and directed evolution. Each dot represents a variant that was selected for its improved $\Delta F/F_0$ in the confirmation assay. HaloKbp0.2, HaloKbp0.5, and HaloKbp0.9 are colored from light to dark purple. The dotted lines separate libraries of different rounds of optimization. (c) Positions of mutations in HaloKbp1a–1c. Colors correspond to text colors used in (a).

affinities for K^+ . Finally, we demonstrated the utility of these indicators for imaging of stimulated changes in the intracellular K^+ concentration.

RESULTS AND DISCUSSION

Rationale for the K^+ Indicator Design. Inspired by the design of the rHCaMP and HaloCaMP series,^{19,20} we decided to use a similar design of protein and the JF₆₃₅-HTL dye (as used for HaloCaMP)²³ to create a K^+ indicator (Figure 1a). The key difference between those previously reported Ca^{2+} indicators and our new K^+ indicator is the use of Kbp rather than a Ca^{2+} -binding domain. Whereas HaloCaMP is based on circularly permuted HaloTag7 with calmodulin fused to one terminus and a Ca^{2+} -binding peptide fused to the other terminus, we elected to use the same topology as rHCaMP (i.e., not circularly permuted) and insert Kbp into HaloTag7. This topology was necessary because Kbp has been reported to not tolerate being split into two polypeptides.¹²

As a rhodamine derivative, JF₆₃₅ fluorophore exists in an equilibrium between a nonfluorescent lactone (L) form and a fluorescent zwitterion (Z) form (Figure 1a, top). This L–Z equilibrium is strongly dependent on the surrounding environment. JF₆₃₅ exists mostly in the lactone form in aqueous solution; however, upon covalent linkage to HaloTag7 (as JF₆₃₅-HTL), the equilibrium partially shifts toward the zwitterionic form, resulting in an increase in fluorescent brightness (i.e., fluorogenic labeling, Figure 1a, top).²³ When attached to HaloTag7, approximately 50% of JF₆₃₅-HTL is in the zwitterionic form, implying that there is opportunity for further on-HaloTag K^+ -dependent modulation of K_{L-Z} .²⁰ We expected that a chimera of HaloTag7 and Kbp could be a chemigenetic K^+ indicator due to the K^+ -binding-induced conformational change altering the L–Z equilibrium of JF₆₃₅ covalently bound to HaloTag7 (Figure 1a). This expected sensing mechanism is conceptually similar to that of single FP-based fluorescent indicators composed of an FP and a target binding protein.²⁹

Identification of a Prototype K^+ Indicator. To realize this design, we first needed to identify a site to insert Kbp into HaloTag7 such that the K^+ -dependent conformational change would modulate the L–Z equilibrium of the attached JF₆₃₅-HTL dye. By manual inspection of the HaloTag7 crystal structure bound to TMR-HTL,²⁰ we identified the region from amino acids 143 to 178 as promising sites that are both solvent-exposed and close to the HTL binding site (Figure 1b, bottom). Accordingly, we created a library of 32 protein variants in which Kbp replaced 2 or more successive amino acids at a range of sites within this region. As linkers between Kbp and HaloTag7, we used GINKO2-derived Pro–Asp as the N-terminal linker (N-linker) and Ala–Ala as the C-terminal linker (C-linker). This design effectively splits HaloTag7 into 2 parts that we refer to as N-Halo and C-Halo (Figure 1c, top). The 32 protein variants, referred to as the HaloKbp series, were individually expressed in *E. coli* and purified, and a 1.5× molar excess of each protein was incubated with JF₆₃₅-HTL. We then tested their fluorescence response ($\Delta F/F_0$, where F_0 refers to fluorescence intensity in the absence of K^+ , F refers to fluorescence intensity in the presence of K^+ , and $\Delta F = F - F_0$) to 100 mM K^+ . Among the tested insertion sites, several of them exhibited positive or negative changes with absolute $\Delta F/F_0 > 0.20$ (Figure 1c, bottom). The variant with the highest $\Delta F/F_0$ value (0.88) was the variant in which Kbp replaced amino acids 145 and 146 of HaloTag7. We designated this variant as HaloKbp0.1₆₃₅ and used it as the prototype for further optimization (Figure 1c, bottom). Notably, this insertion site (at residue 145) is effectively the same as the permutation site used for the development of the HaloCaMP series (at residue 143).²⁰

Optimization of HaloKbp0.1 by Library Screening. The length and composition of the linkers between the binding domain and the fluorescent domain are particularly important for the function of FP-based²⁹ and chemigenetic²⁰ indicators. To optimize the linkers, we first explored the effect of linker length by changing the lengths of both the N-linker and the C-

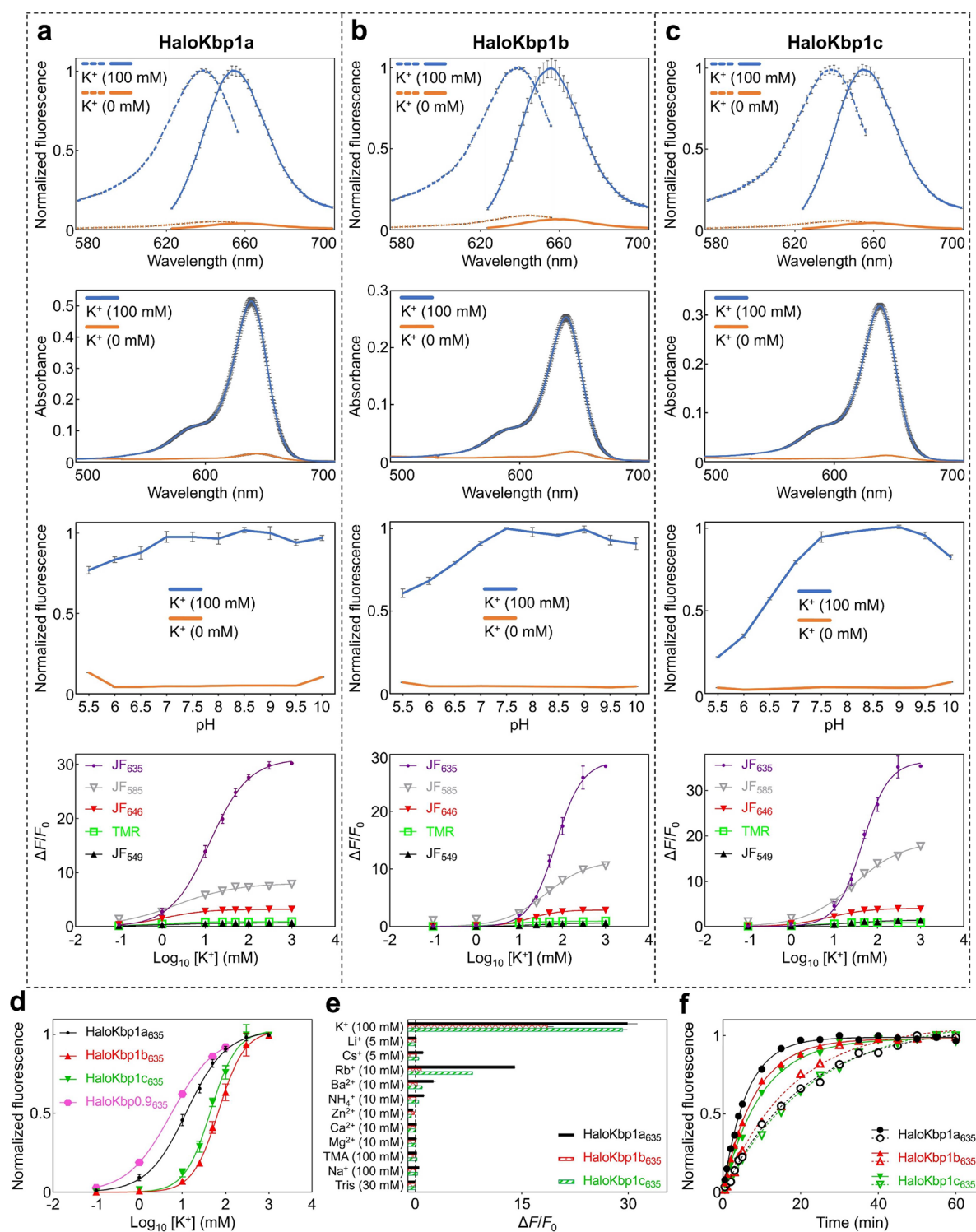


Figure 3. In vitro characterization of the HaloKbp1 series. (a–c) Characterization of HaloKbp1a (a), HaloKbp1b (b), and HaloKbp1c (c). (Top chart) Excitation and emission spectra in the presence (100 mM) and absence of K^+ when labeled with JF₆₃₅-HTL. The dashed line refers to the excitation spectrum, and the solid line refers to the emission spectrum. (Second chart from top) Absorbance spectra in the presence (100 mM) and absence of K^+ when labeled with JF₆₃₅-HTL at a final concentration of 5 μ M. (Third chart from top) pH titration in the presence (100 mM) and absence of K^+ when labeled with JF₆₃₅-HTL. (Bottom chart) K^+ titration curve (K_d measurement) with different rhodamine HTLs. For the top three charts, $n = 3$ technical replicates, and for the K^+ titration charts, $n = 3$ or 4 independent experiments. Mean $\pm$ s.d. (d) Summary of K^+ titration curves (K_d measurements) for HaloKbp0.9 and HaloKbp1 series labeled with JF₆₃₅-HTL ($n = 3$ or 4 independent experiments, mean $\pm$ s.d.). (e) Ion specificity of the HaloKbp1₆₃₅ series. The concentrations of cations used are much higher than their typical physiological concentrations. A dashed line represents no change ($\Delta F/F_0 = 0$). TMA is tetramethylammonium chloride ($n = 3$ technical replicates, mean $\pm$ s.d.). (f) Binding kinetics of the HaloKbp1 series with JF₆₃₅-HTL. The protein concentrations were 2 μ M and the JF₆₃₅-HTL concentrations were 50 nM. Dashed lines refer to measurements in the presence of 100 mM K^+ , and solid lines refer to measurements in the absence of K^+ ($n = 3$ technical replicates).

linker from 0 to 4 amino acids. Shortening the linkers was done by deletion of the two amino acids in each linker, and extension was done by restoring the amino acids that had been deleted in the process of insertion (in this case, A145 and R146 of HaloTag7). After testing all $5 \times 5 = 25$ combinations for their K^+ -dependent $\Delta F/F_0$, we identified a variant with an unchanged N-linker and an extended C-linker (A294-A295-A296-R297) as the best combination. This variant was designated HaloKbp0.2₆₃₅ and exhibited a $\Delta F/F_0$ of 217%. We then optimized the amino acid composition of the linkers by three rounds of library creation by randomizing linker residues, colony-based screening, and analysis of the K^+ -dependent response in protein extracts (Figures 2a,b and S1a). To facilitate colony-based screening, we added JF₆₃₅-HTL to the agar plate and picked colonies based on visual inspection of fluorescence (Figure S1). The best variant after the third round was HaloKbp0.5₆₃₅ with an N-linker of L145-V146 and a C-linker of S294-L295-A296-R297 (Figure 2a), which exhibited a $\Delta F/F_0$ of 835% in response to 100 mM K^+ .

To further improve the performance of HaloKbp0.5₆₃₅, we next performed iterative directed evolution in *E. coli* using error-prone PCR to introduce random mutations throughout the whole gene.³⁰ To date, there have been relatively few examples of using directed evolution to improve the properties of chemigenetic indicators.^{19,20} The screening was performed in a similar manner to linker optimization (Figure S1), but the screened library size was twice as large in each round (2000–4000 colonies over 4–8 Petri dishes). After 7 rounds of directed evolution, we settled on the final variant, which was named HaloKbp0.9₆₃₅ (Figure 2a,b) and exhibited a $\Delta F/F_0$ of 1950% in response to 100 mM K^+ .

Affinity Tuning of HaloKbp0.9₆₃₅ to Create the HaloKbp1 Series. Further characterization of purified HaloKbp0.9₆₃₅ in vitro revealed that its K_d was 5.5 mM. Concerned that this affinity is too high for intracellular applications, we undertook the development of variants with lower affinity for K^+ . We used several strategies including the reversion of specific mutations back to the original amino acid on Kbp, screening of libraries randomized at selected positions (see Supporting Information Methods), and directed evolution with screening for higher response when going from 10 to 100 mM K^+ . These efforts yielded HaloKbp1a₆₃₅ with a K_d of 13 mM (high-affinity variant), HaloKbp1b₆₃₅ with a K_d of 70 mM (low-affinity variant), and HaloKbp1c₆₃₅ with a K_d of 45 mM (medium-affinity variant) (Figures 2a,c and 3a–d).

To further characterize the performance of the HaloKbp1₆₃₅ series (1a, 1b, and 1c) in vitro, we determined the fluorescence spectra, brightness, affinity, $\Delta F/F_0$, kinetics, and pH dependence of the purified proteins reacted with JF₆₃₅-HTL (Figure 3a–c and Table S1). The HaloKbp1₆₃₅ series exhibits significant absorbance, excitation, and emission intensity changes upon binding to K^+ (Figure 3a–c). Specifically, their emission peak at 650 nm (excitation at 550 nm) increases approximately 30-fold in the presence of 1000 mM K^+ (Table S1), with most of the change attributable to changes in the extinction coefficient resulting from a shift in the L–Z equilibrium. All three members of the HaloKbp1₆₃₅ series exhibited relatively stable fluorescence intensity at pH values between 7.5 and 8.5 (Figure 3a–c). The specificity profile of the HaloKbp1 series was practically identical to that of GINKO2,¹³ as expected because both sensors use the Kbp sensing domain (Figure 3e). The only ion that caused a substantial response was Rb^+ , which has a similar ionic radius

to K^+ . However, Rb^+ is unlikely to interfere with K^+ detection due to its low physiological abundance. Rb^+ (at 10 mM for HaloKbp1a₆₃₅ and 1c₆₃₅ and 100 mM for HaloKbp1b₆₃₅) is also the only ion that substantially blocked the K^+ -dependent response of the HaloKbp1₆₃₅ series (Figure S2). Labeling of the HaloKbp1 series with JF₆₃₅-HTL in the absence of K^+ is complete within approximately 40 min, slowing down to approximately 60 min in the presence of 100 mM K^+ (Figure 3f). The rate of labeling with JF₆₃₅-HTL in the absence of K^+ ($k_{app(-)}$) is approximately 3× faster than in the presence of K^+ ($k_{app(+)}$) (Table S1). The HaloKbp1₆₃₅ series all have quantum yields over 0.75 when bound to K^+ , which is higher than that reported for the original JF₆₃₅ dye (0.56).²³ Similar observations have been made for HaloTag7 labeled with JF₆₃₅-HTL,²⁰ HaloCaMP,²⁰ and HaloAKAR,²⁴ which all exhibit quantum yields of 0.7 or greater. We speculate that the optimized protein microenvironment may act to suppress nonradiative relaxation pathways. Measurements of the K^+ binding kinetics revealed k_{on} values of 20, 14, and 28 $M^{-1} \cdot min^{-1}$ and k_{off} values of 0.26, 1.0, and 1.3 min^{-1} for HaloKbp1a₆₃₅, HaloKbp1b₆₃₅, and HaloKbp1c₆₃₅, respectively (Table S1).

Spectral Variants of the HaloKbp1 Series. To determine if the HaloKbp1 series could be used with rhodamine dyes other than JF₆₃₅-HTL, we labeled the proteins with other rhodamine-type HTLs and tested their fluorescence performance for K^+ sensing. We used four HTLs featuring red to far-red emission (TMR-HTL, JF₅₄₉-HTL, JF₅₈₅-HTL, and JF₆₄₆-HTL, Figure S3) and found that all showed K^+ -dependent fluorescence changes, albeit with different levels of fluorescence response (Figure 3a–c, bottom and Table S2). As previously described,²⁰ the K_{L-Z} of the HTLs was found to be inversely correlated with the fluorescence responses (values from refs 20 and 31). Using HaloKbp1c (Figure 3c, bottom) as an example, JF₆₃₅-HTL and JF₅₈₅-HTL with $K_{L-Z} < 0.0001$ had the highest fluorescence response to 1000 mM K^+ ($\Delta F/F_0 = 35.4$ and 17.8, respectively). JF₆₄₆-HTL with $K_{L-Z} = 0.0012$ had a modest fluorescence response to K^+ ($\Delta F/F_0 = 3.85$), and JF₅₄₉-HTL with $K_{L-Z} = 3.5$ and TMR-HTL with $K_{L-Z} = 4.0$ had the lowest fluorescence responses ($\Delta F/F_0 = 1.38$ and 0.820, respectively).³¹ These results suggest that a chemigenetic indicator optimized with one particular dye can also be used with other dyes in the same structural class. We also found that the HaloKbp1₅₈₅ and HaloKbp1₆₃₅ series share similar K^+ binding kinetics with less than 5% difference in their respective k_{on} and k_{off} values (Tables S1 and S2). However, we observed that the apparent affinity of the sensor for the target ion can be dependent on the dye used (Table S2). Similar trends have been observed with the HaloCaMP²⁰ and WHaloCaMP²² series. Our results also provide further support for the conclusion that highly fluorogenic HTL dyes are necessary for the development of high-performance chemigenetic indicators based on shifts in the L–Z equilibrium. Notably, WHaloCaMP was designed to employ a less fluorogenic JF₆₆₉-HTL ligand ($K_{L-Z} = 0.262$)³² that undergoes Ca^{2+} -dependent quenching via photoinduced electron transfer (PET) with tryptophan.²²

Crystal Structure of HaloKbp1a. To better understand the mechanism of the HaloKbp1 series, we determined the crystal structure of HaloKbp1a in the K^+ -bound state. Unfortunately, the structure does not include JF₆₃₅-HTL since high-quality crystals of the HTL-bound state were unattainable (as were crystals of the K^+ -free state).

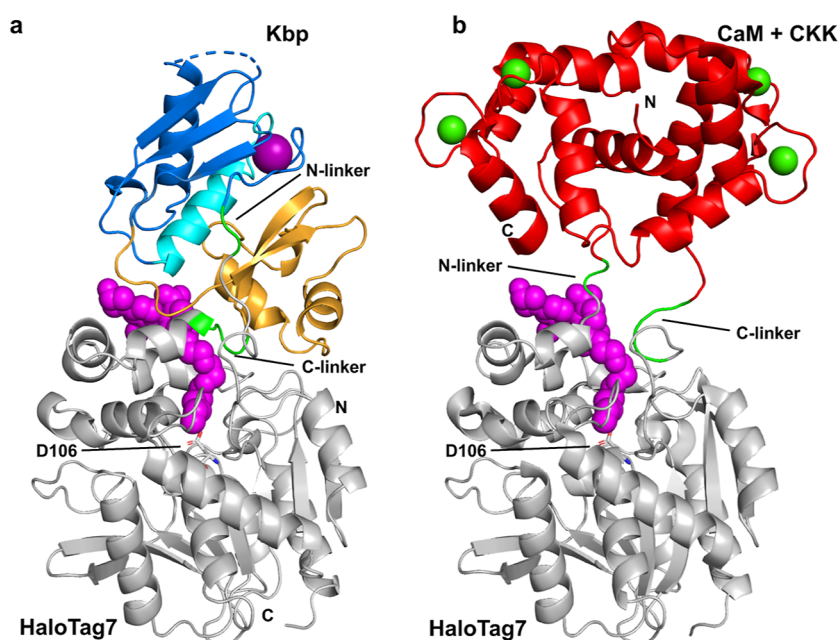


Figure 4. X-ray crystal structure of HaloKbp1a and comparison to HaloCaMP1b. (a) Experimental X-ray structure of HaloKbp1a with a model of bound JF₆₃₅-HTL (i.e., not experimentally determined) represented as magenta spheres. JF₆₃₅-HTL was positioned based on superposition with HaloCaMP1b (PDB ID: 6U2M), as represented in (b).²⁰ D106 (shown in stick representation) is the residue that forms a covalent bond with the HTL. The BON domain is colored blue with its second α -helix colored cyan, the LysM domain is colored orange, the N-linker and C-linker are colored green, and the HaloTag7 domain is colored gray. K⁺ is shown as a purple sphere. (b) Experimental X-ray structure of HaloCaMP1b with bound JF₆₃₅-HTL (PDB ID: 6U2M)²⁰ in the same orientation as HaloKbp1a. CaM and the CKK peptide are colored red, the N-linker and C-linker are colored green, and the HaloTag7 domain is colored gray. Ca²⁺ is shown as green spheres. JF₆₃₅-HTL is represented as magenta spheres. HaloCaMP is based on circularly permuted HaloTag7, but the linker that connects the original C and N termini was not resolved in the crystal structure.

The X-ray crystal structure successfully resolved all amino acids of HaloKbp1a, with the exception of the eight residues from A162 to K169 in the Kbp domain, which is presumably disordered. Both the Kbp domain and the HaloTag7 domain are essentially identical to previously reported structures. The α RMSD for a superposition of the HaloTag7 domain of HaloKbp1a and a previously reported HaloTag7-TMR structure (PDB ID: 6U32)²⁰ was 0.26 Å, and the α RMSD for a superposition of the Kbp domain of HaloKbp1a and the Kbp domain of GINKO1 (PDB ID: 7VCM)¹³ was 0.32 Å. As shown in Figure 4a,b, JF₆₃₅-HTL was positioned in the structure by superposition with the HaloTag7 domain of HaloCaMP1b (a variant of the HaloCaMP series) bound to JF₆₃₅-HTL (PDB ID: 6U2M).²⁰ No additional energy minimization or molecular dynamics was performed on the HaloKbp1a₆₃₅ model, and no major steric clashes between the dye and the protein were apparent. An important caveat is that the JF₆₃₅-HTL has some conformational flexibility, and the dye portion may only partially occupy this position or reside in a different position altogether.

Overall, the structure of HaloKbp1a is similar to that of HaloCaMP1b in terms of the position and orientation of the ion-binding domain (CaM + CKK peptide and Kbp) relative to the HaloTag7 domain (Figure 4a,b). This is to be expected, given the fact that the permutation site of HaloCaMP1b (where CaM + CKK peptide are fused to the termini) and insertion site of HaloKbp (where Kbp is inserted into HaloTag7) are around positions 143 and 145, respectively. The interface areas between the HaloTag7 domain and the ion-binding domains in HaloKbp1a and HaloCaMP1b are 1140 Å² for HaloKbp1a and 447.5 Å² for HaloCaMP1b

(calculated using PISA).³³ Moreover, 14 hydrogen bonds are formed across the interface in HaloKbp1a, compared to 3 for HaloCaMP1b. This demonstrates that the Kbp and HaloTag7 domains of HaloKbp1a are more intimately associated than the CaM + CKK peptide and HaloTag7 domains of HaloCaMP1b (Figure 4a,b). Relative to HaloCaMP1b ($\Delta F/F_0 = 9$),²⁰ the closer association of the ion-binding domain and HaloTag7 domain in HaloKbp1a ($\Delta F/F_0 = 30$) may have made the protein more amenable to optimization of the brightness of the bound state and $\Delta F/F_0$, via directed evolution.

Due to the absence of the critical JF₆₃₅-HTL in the experimentally determined structure, we are hesitant to assign specific mechanistic roles to any specific mutations acquired during the development of HaloKbp1a (Figure S4). However, we do note that in the model with the superpositioned JF₆₃₅-HTL, the second α -helix of the BON domain (highlighted in cyan in Figure 4a) is positioned with K⁺ near its C-terminal end and the anionic JF₆₃₅ carboxylate near its N-terminal end. We tentatively speculate that the inherent dipole of this helix might be acting as an allosteric or electrostatic bridge through which K⁺ binding could modulate the L–Z equilibrium of JF₆₃₅-HTL.

Satisfyingly, we found that the structure of the Kbp domain in HaloKbp1a predicted by AlphaFold2 (refs 34 and 35), which we had used to guide our protein engineering efforts, was in reasonably good agreement with the X-ray crystal structure (Figure S5). The recently released AlphaFold3 (ref 36), which has the capability to model K⁺ ions, predicted a structure in which the Kbp and HaloTag7 domains were essentially identical to the X-ray crystal structure (Figure S6a,b) but substantially different in terms of their relative orientation

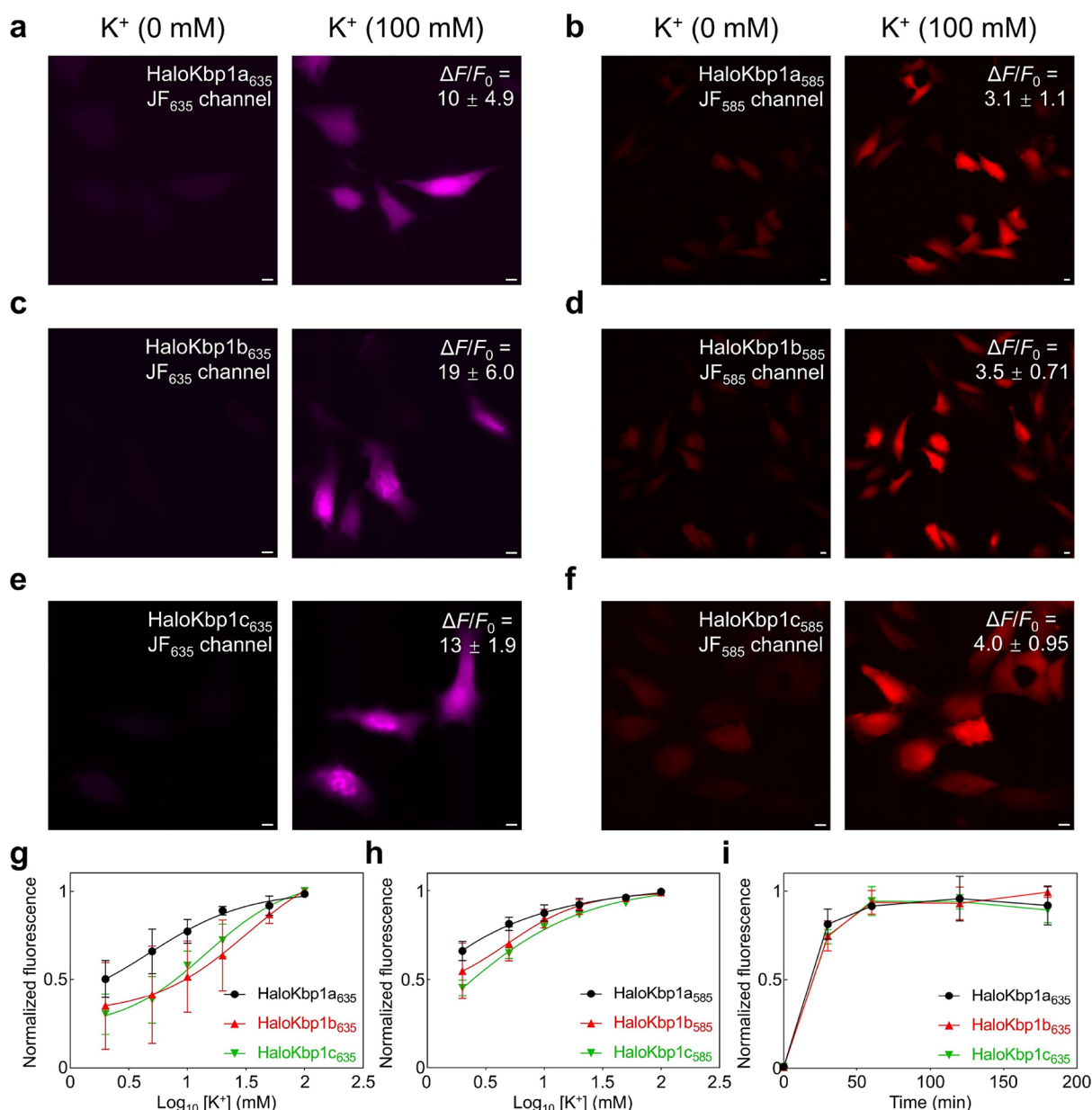


Figure 5. Cell-based characterization and application of the HaloKbp1 series. (a–f) K⁺ intracellular imaging of HeLa cells expressing HaloKbp1a (a,b), HaloKbp1b (c,d), and HaloKbp1c (e,f), fully labeled with JF₆₃₅-HTL (a,c,e) or JF₅₈₅-HTL (b,d,f) in the 670 nm channel (JF₆₃₅-HTL channel) or the 600 nm channel (JF₅₈₅-HTL channel). Scale bar = 10 μ m. For each panel, the left images were taken in K⁺ (0 mM) buffer, while the right ones were taken in K⁺ (100 mM) buffer. Cells were treated with 4 μ M digitonin before the measurement to remove the endogenous K⁺. F and F_0 are determined as average fluorescence intensities of the cells shown above in K⁺ (100 mM) buffer and K⁺ (0 mM) buffer, respectively, and $\Delta F = F - F_0$. (g) Intracellular K⁺ titration dose–response curve (K_d measurement) of the HaloKbp1₆₃₅ series. HaloKbp1a₆₃₅ = (63,4), HaloKbp1b₆₃₅ = (36,3), and HaloKbp1c₆₃₅ = (43,3), where (x,n) = (number of cells in total which were selected as the plot area obtaining fluorescence data, number of independent experiments in independent transfections on different days). Mean $\pm$ s.d. (h) Intracellular K⁺ titration dose–response curve (K_d measurement) of the HaloKbp1₅₈₅ series. HaloKbp1a₅₈₅ = (48,3), HaloKbp1b₅₈₅ = (42,3), and HaloKbp1c₅₈₅ = (44,3), where (x,n) are the same as in the legend for (g). Mean $\pm$ s.d. (i) Binding kinetics of the HaloKbp1 series expressed in intact HeLa cells when binding to JF₆₃₅-HTL. In imaging buffer, the final concentration of JF₆₃₅-HTL is 1 μ M without adding any other reagents ($n = 3$ independent experiments in independent transfections, mean $\pm$ s.d.).

(Figure S6c). When we attempted to position JF₆₃₅-HTL in the five AlphaFold3-predicted structures by superposition with the HaloTag7 domain of HaloCaMP1b (as we had done with the experimental structure), we found that there was a steric clash with the Kbp domains (Figure S6d).

Intracellular Characterization of the HaloKbp1 Series.

To perform mammalian cell imaging experiments using the HaloKbp1 series, we transiently expressed the HaloKbp1

proteins in the cytosol of HeLa cells and then treated the cells with JF₆₃₅-HTL (1 μ M, final concentration). To assess whether the indicators retained their K⁺-responsiveness under these conditions, we permeabilized the cells with digitonin to deplete K⁺ from the cytosol and then added K⁺ at known concentrations. Under these conditions, the HaloKbp1₆₃₅ series underwent a $\Delta F/F_0$ of 10 ± 4.9 for HaloKbp1a₆₃₅, 19 ± 6.0 for HaloKbp1b₆₃₅, and 13 ± 1.9 for HaloKbp1c₆₃₅ (for

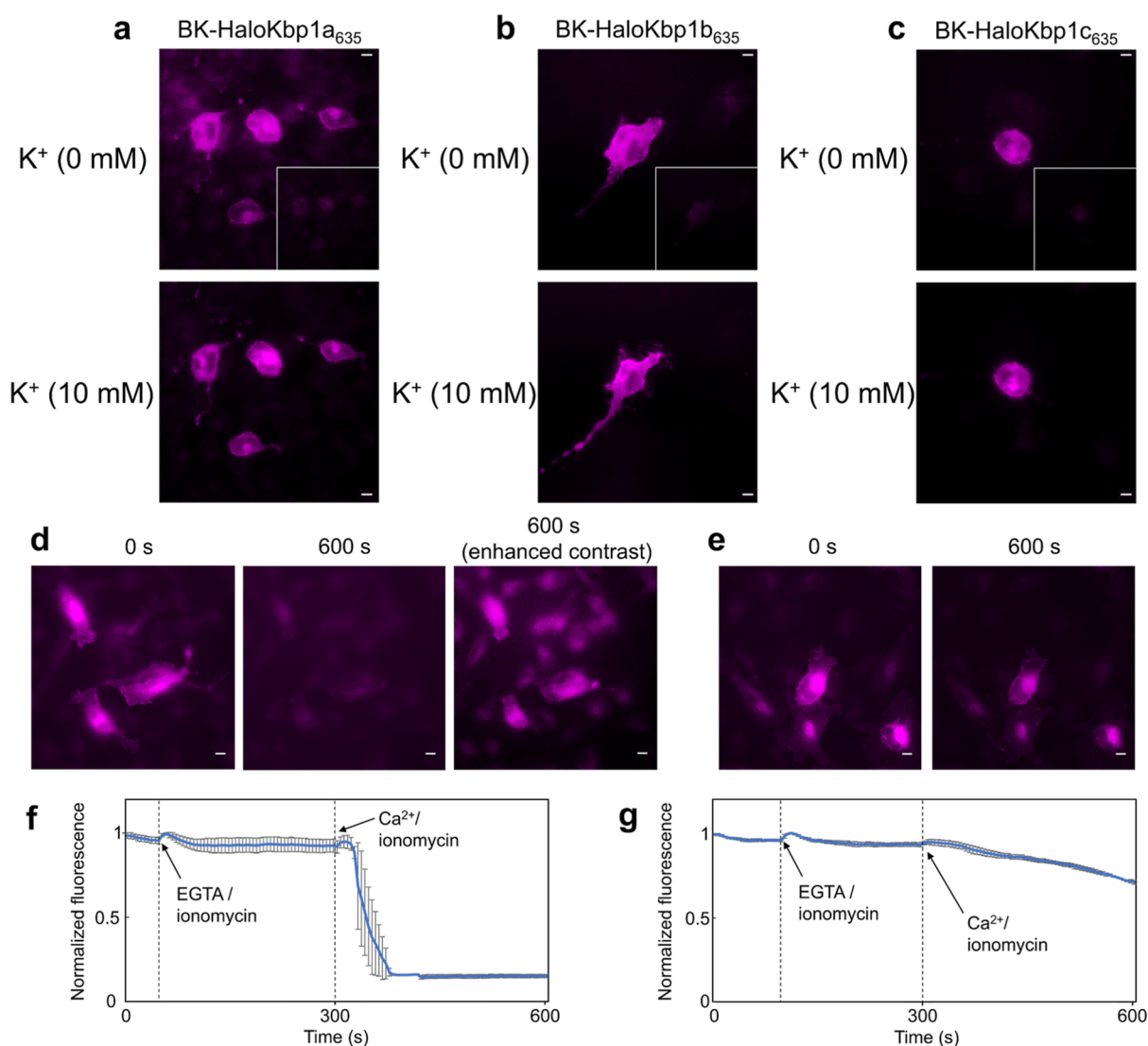


Figure 6. Application of the BK-HaloKbp1₆₃₅ series to monitor cellular changes in K⁺ concentration. (a–c) K⁺-dependent fluorescence response of BK-HaloKbp1a₆₃₅ (a), BK-HaloKbp1b₆₃₅ (b), and BK-HaloKbp1c₆₃₅ (c). For the K⁺ (0 mM) images, the main images have enhanced contrast, while the inset images have the same brightness/contrast settings as the K⁺ (10 mM) images. Cells were treated with 4 μ M digitonin before the measurement. Similar trends were observed for >30 cells in two or three additional experiments following independent transfections. Scale bar = 10 μ m. (d) Imaging of HeLa cells expressing BK-HaloKbp1a₆₃₅ before and after Ca²⁺ stimulation. Scale bar = 10 μ m. (e) Imaging of HeLa cells expressing BK-HaloKbp1a₆₃₅ before and after Ca²⁺ stimulation with 10 nM iberiotoxin pretreatment. Scale bar = 10 μ m. (f) Ca²⁺-triggered K⁺ efflux from cells expressing BK-HaloKbp1a₆₃₅. Fluorescence is determined as the average fluorescence intensity of three of the cells shown in (d) ($n = 3$ cells in this individual experiment, mean $\pm$ s.d.). Similar trends were monitored in two or three additional experiments carried out in the same manner in independent transfections. (g) Ca²⁺-triggered K⁺ efflux from cells expressing BK-HaloKbp1a₆₃₅ with 10 nM iberiotoxin pretreatment. Fluorescence is determined as the average fluorescence intensity of four of the cells shown in (e) ($n = 4$ cells in this individual experiment, mean $\pm$ s.d.). Similar trends were monitored in two or three additional experiments carried out in the same manner in independent transfections.

the cells in Figure 5a,c,e), in good agreement with the results obtained with purified proteins. Treatment with JF₆₃₅-HTL instead of JF₅₈₅-HTL resulted in $\Delta F/F_0$ values of 3.1 ± 1.1 for HaloKbp1a₅₈₅, 3.5 ± 0.71 for HaloKbp1b₅₈₅, and 4.0 ± 0.95 for HaloKbp1c₅₈₅ (for the cells in Figure 5b,d,f). Treatment with TMR-HTL resulted in a $\Delta F/F_0$ of 0.68 ± 0.21 (Figure S7), which is in line with the in vitro results using rhodamine-type ligands with different fluorogenicities.

In addition to the maximum K⁺-dependent fluorescent response, we also assessed the affinity, labeling kinetics, and response kinetics in cells. The intracellular K_d values were determined to be lower than the values with purified proteins for both JF₆₃₅-HTL and JF₅₈₅-HTL (Figure Sg,h and Tables S1 and S2), as has been previously reported for other indicators

based on Kbp.^{13,14} Labeling of the HaloKbp1 series with JF₆₃₅-HTL was complete within 60 min (Figure 5i). This time scale is similar to that reported for HaloTag7 labeling with rhodamine HTLs.^{31,37} Upon the addition of 100 mM K⁺, the fluorescence increased to its maximum within about 60 s (Figure S8).

Monitoring of K⁺ Efflux Using the BK-HaloKbp1₆₃₅ Series. In an effort to assess the utility of the HaloKbp1₆₃₅ series, we attempted to use it to monitor K⁺ efflux through the human BK channel. The BK channel is a Ca²⁺-activatable K⁺ channel that has its N-terminus on the extracellular side of the cell membrane and its C-terminus on the intracellular side of the cell membrane.³⁸ Accordingly, theoretically, it is possible to localize the HaloKbp1 protein on either the inner or outer side

of the plasma membrane (Figure S9) by fusing it to the C- or N-terminal end, respectively, of the BK channel.

We found that cells expressing HaloKbp1-BK (i.e., with HaloKbp1 on the extracellular side) did not exhibit any fluorescence when treated with JF₆₃₅-HTL. In an additional effort to get a functional HaloKbp1₆₃₅ on the extracellular side of the membrane, we explored using a CD59 membrane-targeting leader³⁹ and a variety of membrane-targeting anchors⁴⁰ (Figure S10a). We succeeded in targeting the protein to the extracellular side, as indicated by strong fluorescence labeling. However, we were unable to do so in a way that retained the indicator function (Figure S10b), consistent with our previous experiences at attempting to target the GINKO2 K⁺ sensor to the extracellular side of the membrane.¹³ In contrast, the BK-HaloKbp1₆₃₅ series (i.e., with HaloKbp1 on the intracellular side) was well-expressed and functional. Cells expressing the BK channel often appeared somewhat more rounded than usual. When we permeabilized HeLa cells and modulated the K⁺ concentration, the BK-HaloKbp1₆₃₅ series exhibited a substantial fluorescence response to K⁺ (Figure 6a–c).

To demonstrate the applicability of the BK-HaloKbp1₆₃₅ series for both inducing and monitoring K⁺ efflux triggered by Ca²⁺, we added Ca²⁺ plus ionomycin to the media while imaging the fluorescence of the BK-HaloKbp1₆₃₅ series (schematically represented in Figure S11). Cells were first treated with EGTA plus ionomycin to remove endogenous Ca²⁺. Upon addition of Ca²⁺/ionomycin, BK-HaloKbp1a₆₃₅ exhibited an 85% decrease in fluorescence intensity (Figure 6d,f), with BK-HaloKbp1b₆₃₅ and BK-HaloKbp1c₆₃₅ exhibiting similar trends (Figure S12). Control experiments with cytosolic JF₆₃₅-HTL-labeled HaloTag7 revealed no substantial fluorescence changes (Figure S13). Incubating cells with iberiotoxin⁴¹ (a specific BK channel inhibitor) prior to the Ca²⁺ stimulus (schematically represented in Figures S14) greatly suppressed the fluorescence decrease compared to the nontreated cells (Figures 6e,g and S15), consistent with the fluorescence decrease being triggered by the opening of the BK channel and K⁺ efflux. A gradual decrease in fluorescence following Ca²⁺/ionomycin treatment, which could not be explained by photobleaching or cell movement, was consistently observed. Possible explanations are incomplete inhibition of the BK channel and ionomycin-mediated transport of K⁺. Although ionomycin has very little affinity for monovalent cations,⁴² we speculate that some binding and transport may occur at the very high K⁺ concentration present in cells. An additional control experiment by treating the cells with NS1619 (ref 43) as the BK channel opener (schematically represented in Figure S16) revealed a decrease of fluorescence comparable to that obtained with the Ca²⁺ treatment (Figure S17). Overall, these results demonstrate that the BK-HaloKbp1₆₃₅ series can be used to monitor K⁺ efflux resulting from the opening of the BK channel.

Mitochondria Targeting of the HaloKbp₆₃₅ Series.

Recent studies have demonstrated that K⁺ fluxes through a mitochondrial membrane have important effects on inflammation formation,⁴⁴ insulin secretion,⁴⁵ and apoptosis.⁴⁶ Although there have been reports of mitochondrially localized K⁺ indicators,^{47,48} they are synthetic and unsuitable for in vivo applications. We expected that the HaloKbp1₆₃₅ series could provide a powerful means to monitor mitochondrial K⁺. To explore this possibility, we first demonstrated that HaloKbp₆₃₅ could be targeted to either the cytosolic face of the outer

mitochondrial membrane (Figure S18) or the mitochondrial matrix (Figure S19) while maintaining the fluorescence response to the changes in K⁺ concentration. We noticed that the indicators expressed in the mitochondrial matrix had a diminished dynamic range compared to indicators targeted to the mitochondrial membrane and cytosol (for example, HaloKbp1a₆₃₅ gave $\Delta F/F_0$ of 5.4 ± 0.9 in the mitochondrial matrix and $\Delta F/F_0$ of 9.5 ± 1.1 on the outer mitochondrial membrane upon treatment with 10 mM K⁺). However, this could be partially due to the imperfect permeabilization of the mitochondrial membrane, resulting in a residual fluorescence signal without K⁺ addition.

To achieve simultaneous imaging of cytosolic and mitochondrial K⁺, we coexpressed HaloKbp1 in the mitochondrial matrix and GINKO2 in the cytosol and added K⁺ to permeabilized HeLa cells. We observed fluorescence increases in both channels, further demonstrating the functionality of the HaloKbp1₆₃₅ series inside mitochondria as well as their compatibility to multiplex with green-emitting fluorescent indicators (Figure S20). In both the GINKO2 channel and the JF₆₃₅ channel, we observed a decreased fluorescence change compared to their respective individual expression.¹³ We speculate that this might be due to buffering or cytotoxicity associated with an increased level of protein expression.

CONCLUSIONS

We have reported a series of high-performance chemigenetic K⁺ indicators designated as the HaloKbp1 series. Relative to previously reported genetically encoded^{12–14} and chemigenetic^{26,49,50} K⁺ indicators, the HaloKbp1 series has a number of important advantages that should make them one of the preferred tools of imaging intracellular K⁺ concentrations. Specifically, the HaloKbp1 series, when labeled with the JF₆₃₅ dye featuring far-red emission, exhibits >25× intensity changes in response to K⁺, a brightness that is ~4× that of the brightest green-emitting fluorescent genetically encoded K⁺ biosensor,¹³ and includes affinity variants with K_d values ranging from 13 to 70 mM (Table S3). This series could potentially be applicable to a range of intracellular K⁺ imaging applications, either alone or spectrally multiplexed, as demonstrated in this work by monitoring induced changes in BK channel activity and mitochondrial K⁺ concentration. However, we recognize that most physiologically relevant changes in the K⁺-concentration tend to be much lower than the artificially induced changes used in this work. While the HaloKbp1 series represents the highest performance K⁺ indicators currently available, the full scope of suitable applications for genetically encoded and chemigenetic K⁺ indicators remains to be determined.

There are some limitations to the HaloKbp1 series that may be overcome in the next generation of these indicators. One limitation is that the kinetics of the K⁺-induced fluorescence increase are substantially slower than those for other indicators. For example, GINKO2 has a k_{on} that is about 3× faster and a k_{off} that is about 40% faster than that of HaloKbp1c.¹³ A similar trend is observed when we compare the kinetics of HaloCaMP²⁰ and GCaMP6s,⁵¹ suggesting that it may be a general weakness of chemigenetic indicators. In addition, as with other Kbp-based K⁺ indicators such as GINKO2,¹³ we were unable to localize fully functional HaloKbp1 on the extracellular side of the plasma membrane. In contrast, we were able to localize the functional indicator at the inner plasma membrane and on the mitochondrial outer membrane, indicating that proximity to the membrane itself is

not the reason for this loss of function. While we are currently unable to explain this result, we suspect that there are specific proteins, glycans, or lipid headgroups on the surface of the plasma membrane that disrupt the function of the Kbp protein. Addressing this problem might require exploration of a wider range of natural K⁺-binding proteins.

Some of the other limitations of the HaloKbp1 indicators are those that could affect essentially all chemigenetic indicators.⁵² Although background fluorescence signal from free ligands can be averted when applying highly fluorogenic ligands with low K_{L-Z} values, these same dyes are also associated with decreased in vivo bioavailability relative to dyes with higher K_{L-Z} values.⁵³ For in vivo applications, we may need to use an alternative dye and sensing mechanism such as those reported for WHaloCaMP.²² Although the HaloKbp1 series retained function when highly fluorogenic JF₆₃₅ was swapped for other similar dyes (i.e., highly fluorogenic JF₅₈₅ and less fluorogenic JF₆₄₆), the performance was diminished. To achieve higher performance with less fluorogenic dyes, optimization of the indicator with a specific dye would likely be necessary.

The HaloKbp1 series represents the latest addition to the small but growing family of HaloTag-based chemigenetic indicators for biological imaging. Our success at developing the HaloKbp1 series provides further support for the idea that the HaloTag-dye combination is comparable to FPs, in terms of versatility and generalizability, for the construction of fluorescent indicators using protein-based recognition domains.²⁹ Our work also demonstrates that the process of optimizing chemigenetic indicators can proceed analogously to the established process of optimizing FP-based indicators (i.e., insertion site screening, linker optimization, and iterative directed evolution) with similar expectations of effectiveness.

■ ASSOCIATED CONTENT

Data Availability Statement

Plasmids encoding HaloKbp1a, HaloKbp1b and HaloKbp1c have been deposited in Addgene (#228793, #228794, and #228795). The HaloKbp1a X-ray crystal structure (without JF₆₃₅-HTL) has been deposited in the Protein Data Bank (PDB ID: 8ZEX).

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/jacs.4c10917>.

Directed evolution scheme, additional ion specificity properties, chemical structure of additional rhodamine HTLs, mutations accumulated during protein optimization, structural alignment of the Kbp domain and HaloKbp1a X-ray crystal structure, imaging of intracellular K⁺ in HeLa cells, time course of fluorescence increase, schematic representation of HaloKbp, attempted localization of HaloKbp1, schematic illustration of the Ca²⁺-induced K⁺ efflux experiment, fluorescence imaging of K⁺ efflux, control experiment performed using HeLa cells expressing HaloTag7, schematic illustration of the experiment using BK-HaloKbp1₆₃₅ and iberiotoxin, schematic illustration of the experiments using BK-HaloKbp1₆₃₅ and NS1619, control experiment using the NS1619 BK channel opener to induce K⁺ efflux, imaging of the localization of the HaloKbp1₆₃₅ series, multiplexed imaging of the HaloKbp1₆₃₅ series, NMR spectra, fluorescent properties, comparison of the HaloKbp1₆₃₅ series with existent high-performance

genetically encoded K⁺ indicators, X-ray data collection and refinement statistics, DNA sequences, protein sequences, and methods (general materials and reagents, synthesis, protein engineering methods, in vitro characterization, crystal structure determination, and cell imaging) (PDF)

■ AUTHOR INFORMATION

Corresponding Authors

Takuya Terai – Department of Chemistry, Graduate School of Science, The University of Tokyo, Bunkyo-ku, Tokyo 113-0033, Japan; Email: terai@chem.s.u-tokyo.ac.jp

Robert E. Campbell – Department of Chemistry, Graduate School of Science, The University of Tokyo, Bunkyo-ku, Tokyo 113-0033, Japan; CERVO Brain Research Center and Department of Biochemistry, Microbiology, and Bioinformatics, Université Laval, Québec, Québec G1V 0A6, Canada; orcid.org/0000-0003-0604-092X; Email: campbell@chem.s.u-tokyo.ac.jp

Authors

Dazhou Cheng – Department of Chemistry, Graduate School of Science, The University of Tokyo, Bunkyo-ku, Tokyo 113-0033, Japan

Zhenlin Ouyang – Center for Microbiome Research of MedX Institute, The First Affiliated Hospital, Xi'an Jiaotong University, Xi'an, Shaanxi 710061, China

Xiaoyu He – The Key Laboratory of Environment and Genes Related to Disease of Ministry of Education Health Science Center, Xi'an Jiaotong University, Xi'an, Shaanxi 710061, China

Yusuke Nasu – Department of Chemistry, Graduate School of Science, The University of Tokyo, Bunkyo-ku, Tokyo 113-0033, Japan; PRESTO, Japan Science and Technology Agency, Chiyoda-ku, Tokyo 102-0075, Japan; Institute of Biological Chemistry, Academia Sinica, Nankang, Taipei 115, Taiwan; orcid.org/0000-0002-3193-9104

Yurong Wen – Center for Microbiome Research of MedX Institute, The First Affiliated Hospital, Xi'an Jiaotong University, Xi'an, Shaanxi 710061, China

Complete contact information is available at: <https://pubs.acs.org/doi/10.1021/jacs.4c10917>

Notes

The authors declare no competing financial interest.

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